

INVENTORY & STOCKOUT RISK ANALYSIS

An Inventory Stock Analysis to Identify Stock Risk Across Agrovets, Pharmacies, Supermarkets, and Electronics Shops

Prepared for: Business Stakeholder / Ops Lead

Basis: Engineered dataset (5,526 records, 20 products, 4 business types) — illustrative, not live business records

Prepared by: Evalyn Njagi

Executive Summary

The headline: stock is leaking money, mainly through one business type and one category

KES 4.36M

Total shrinkage loss
recorded

8.9%

of revenue lost
to shrinkage

2,918

stockout-risk
events logged

99.9%

ROC-AUC of model
built to flag risk early

What the data shows

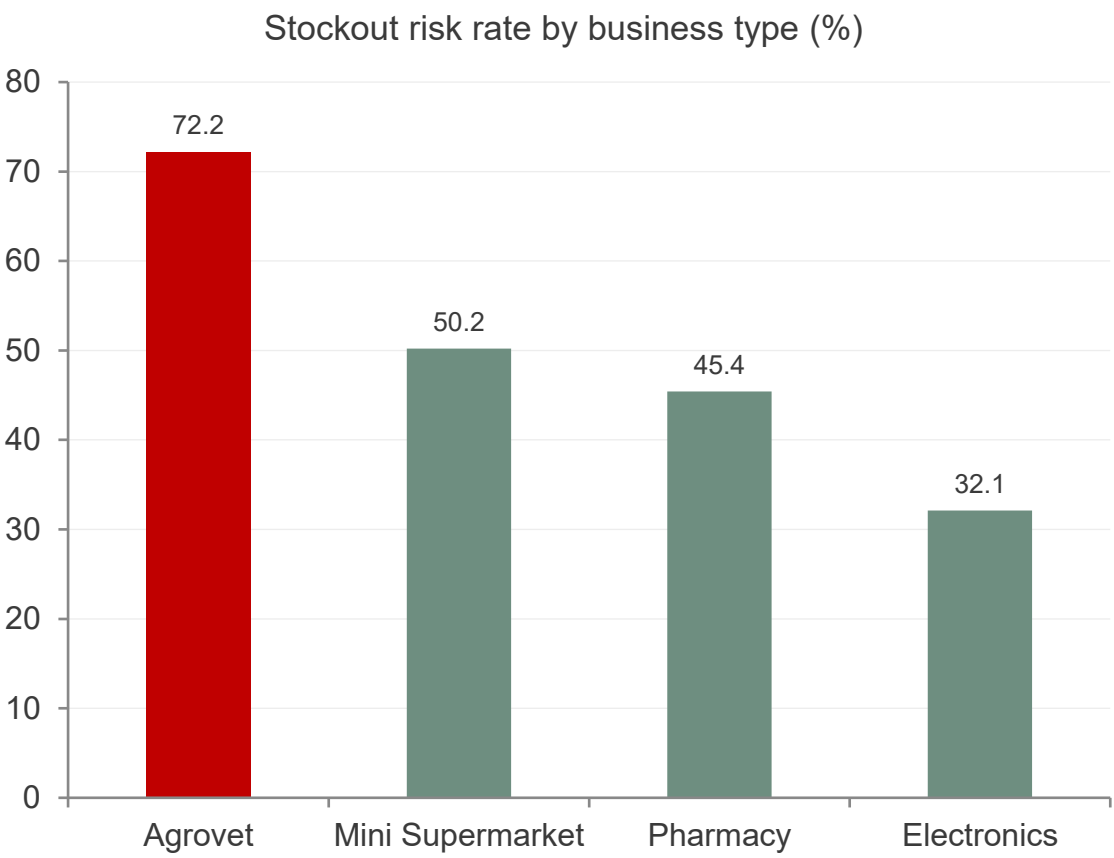
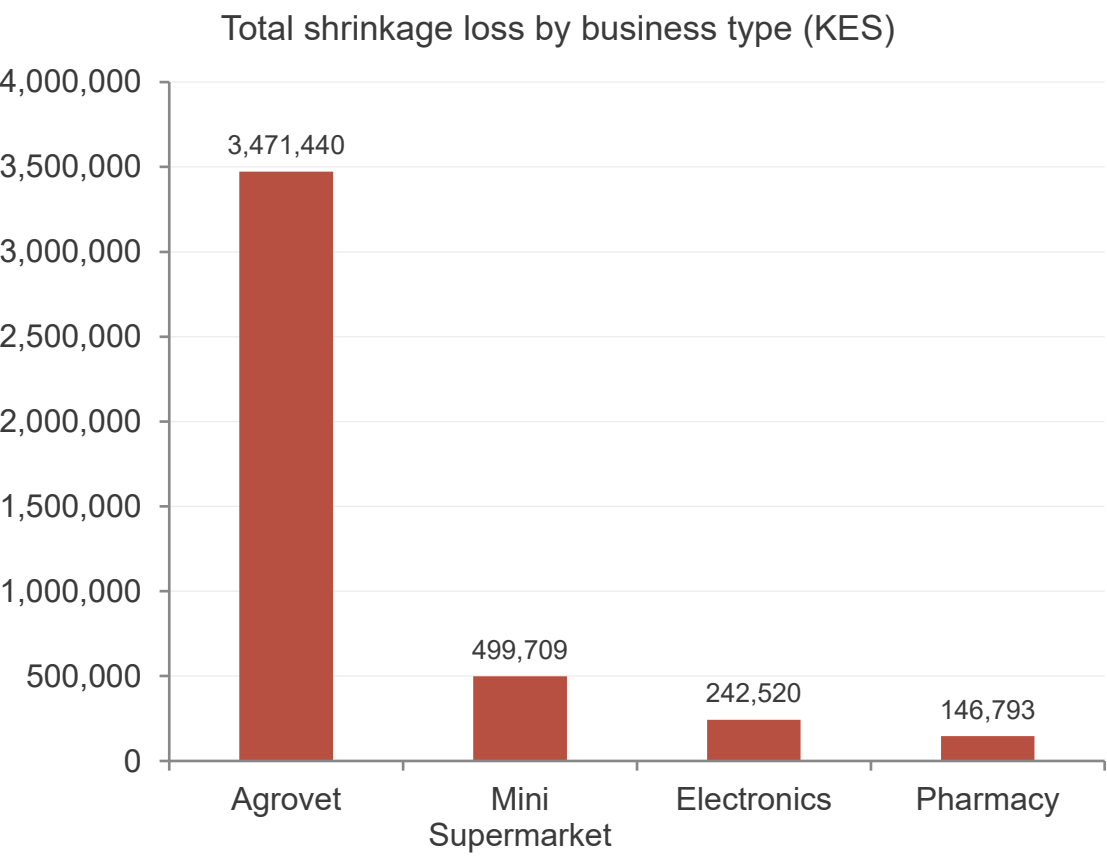
- Agrovets carry the most pain: highest shrinkage value (KES 3.47M) and the highest stockout-risk rate (72.2% of days).
- Fertilisers are the single biggest leak: KES 3.0M of the KES 4.36M total shrinkage loss (~69%).
- Stock levels ,not seasonality or pay-day timing ,are the strongest predictors of stockouts.
- A model trained on this data can flag at-risk stock with 98% precision and recall, well ahead of a stockout.

Recommended action

Prioritize agrovet fertiliser stock: tighten reorder points on DAP and CAN Fertilizer and pilot the risk-prediction model on agrovet locations first before rolling out chain-wide.

The Problem: Current Situation

Two linked issues: shrinkage (loss) and stockouts (missed sales) - hit some business types far harder than others



Combined, agrovets sit at the top of both charts: the most money lost AND the highest share of days running low on stock - the clearest place to intervene first.

Data Overview & Key KPIs

Scope of the analysis and the top-line numbers behind it

Dataset scope

Records analyzed	5,526 daily stock entries
Products tracked	20 products across 15 categories
Business types	Agrovet, Mini Supermarket, Pharmacy, Electronics
Suppliers	14 named suppliers, 4 counties
Time span	Jan 2024 – Nov 2026 (daily)

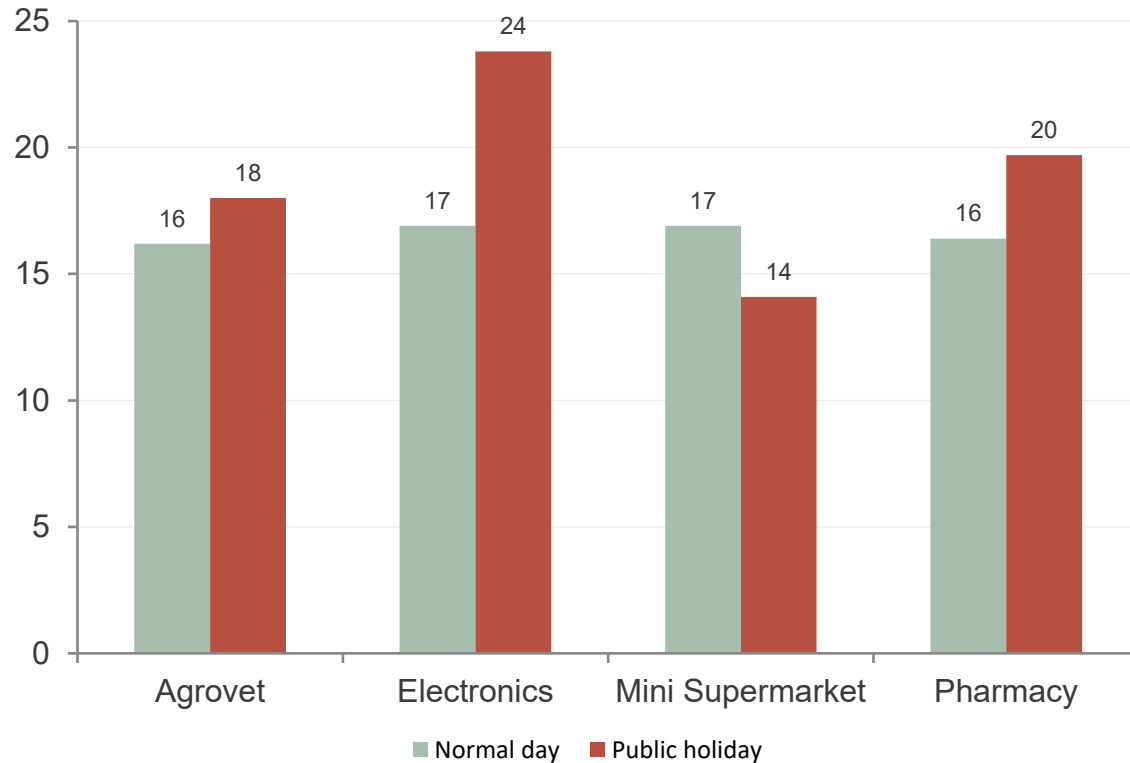
[uncertain] Dataset is a synthetically engineered file built to mirror realistic SME conditions (seasonality, data-quality issues, category shrinkage patterns) - figures below illustrate the method, not verified live sales.

KPI	Value
Total revenue	KES 48,803,617
Total shrinkage loss	KES 4,360,462
Shrinkage as % of revenue	8.93%
Average gross margin	30.1%
Stockout risk events	2,918 product-days
Dead-stock products (zero sales)	0
Products needing reorder now (<2 days stock)	5 (Critical)

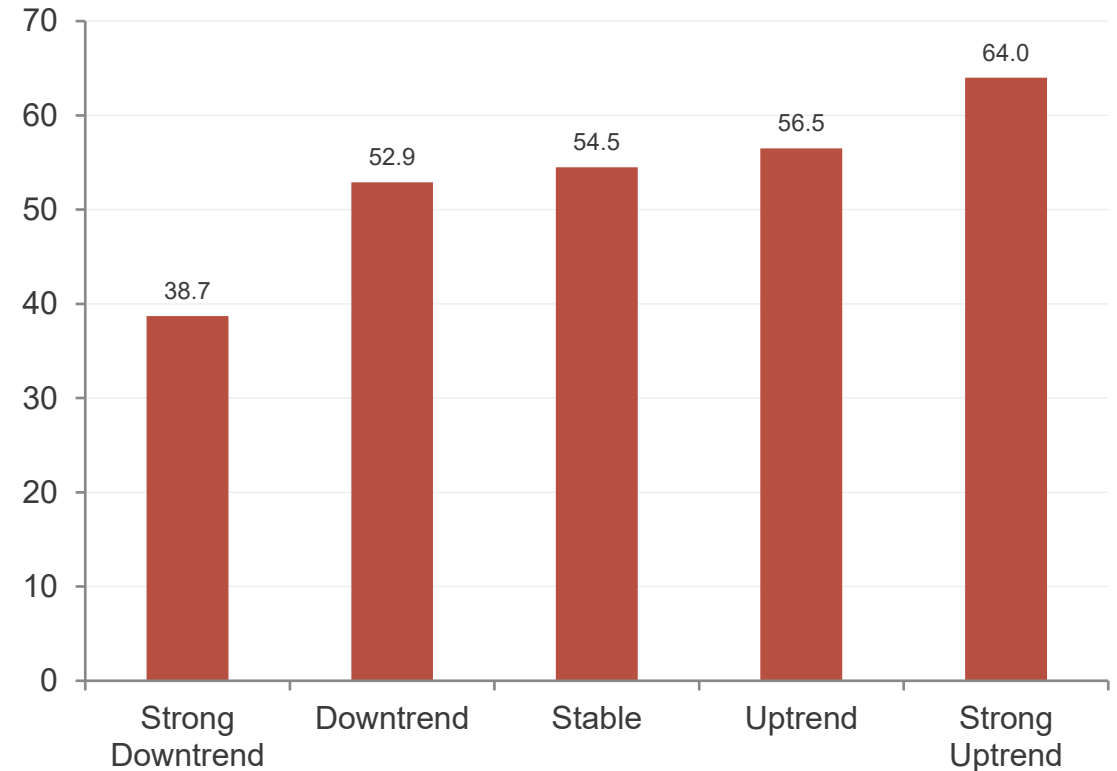
The Data Story: Trends & Insights

What drives demand and risk over time

Avg units sold: normal day vs public holiday



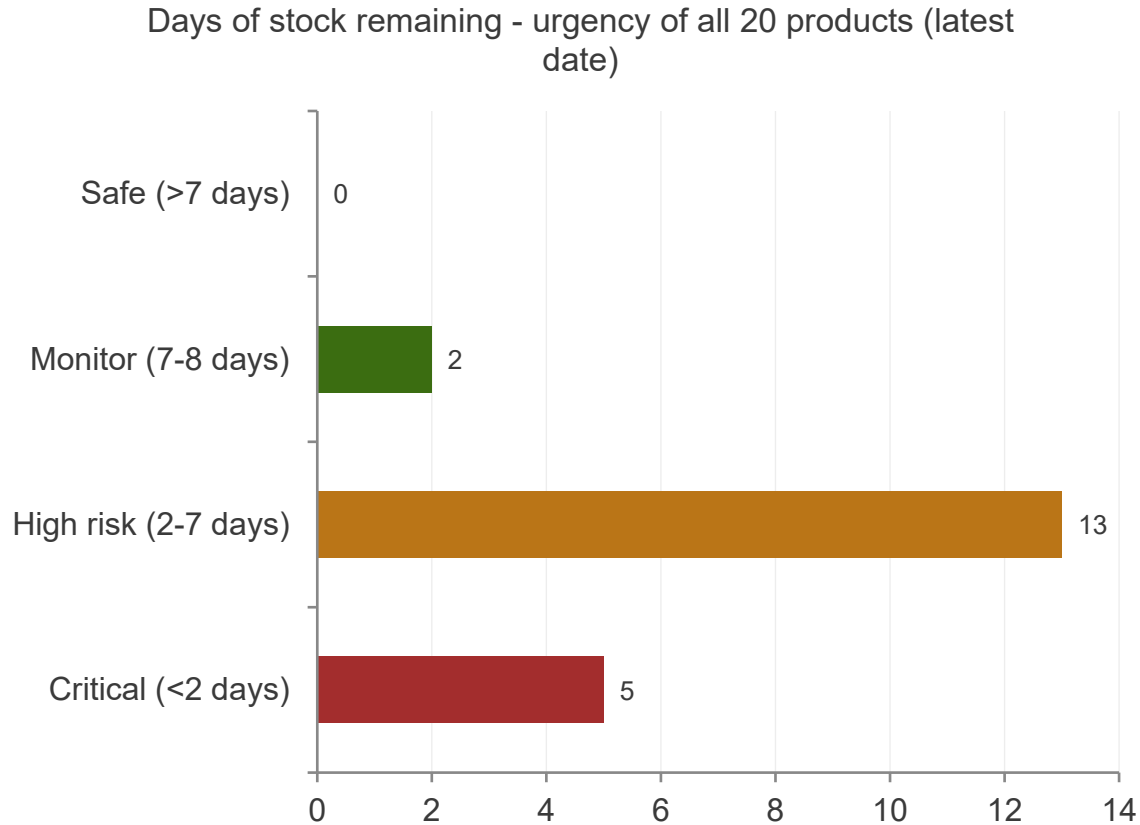
Stockout risk rises with demand trend



- Electronics sees a sharp holiday demand spike (+41%); mini supermarkets actually dip on holidays.
- Products already trending up in sales are, counter-intuitively, at higher stockout risk (64% of days) - fast-rising demand outruns reorder points.
- [uncertain] Month-end (pay-day) demand and Kenyan planting-season fertiliser demand both showed only a small lift (~1% and a moderate seasonal bump respectively) - weaker seasonal signals than expected.

Identified Gaps

Where current practice is falling short, product by product

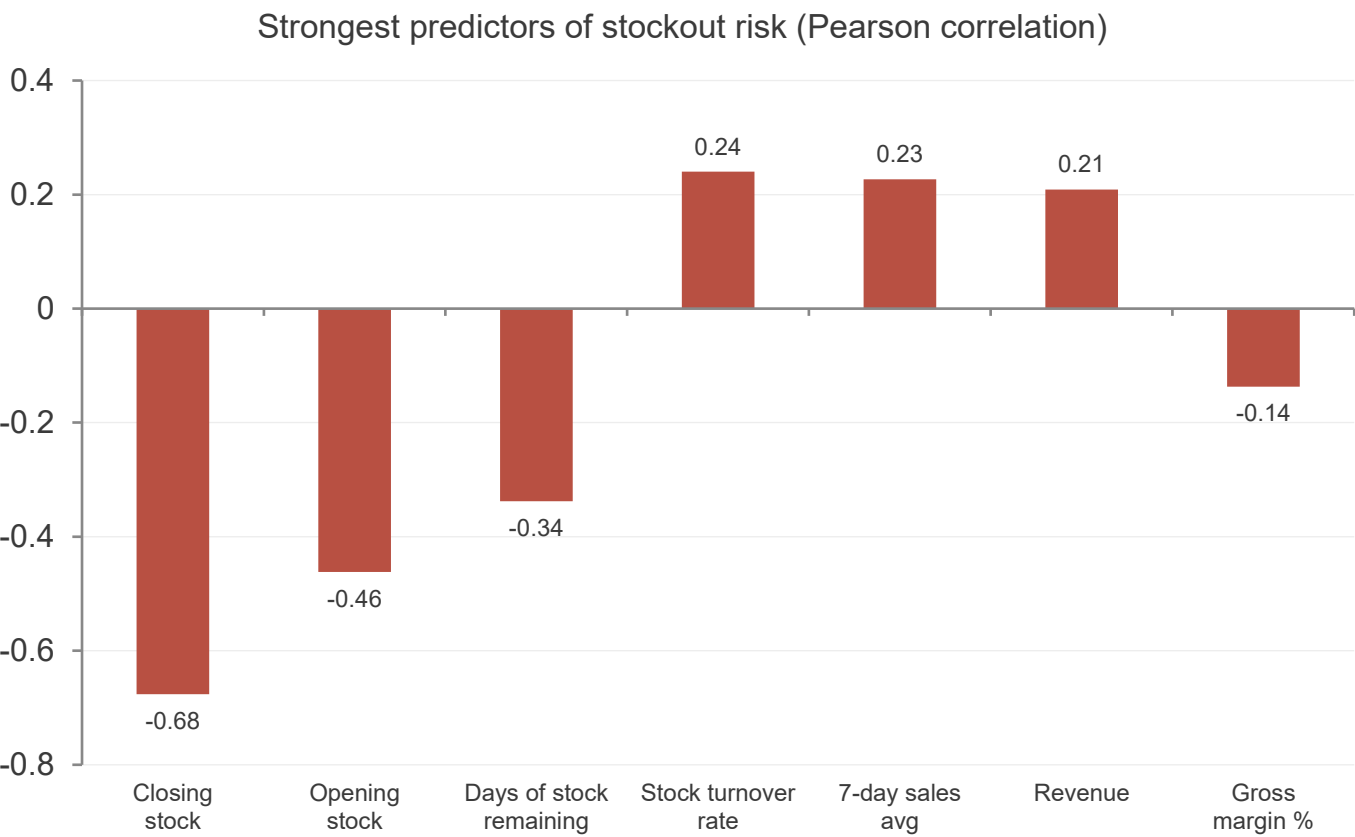


Reading this chart

- 18 of 20 products (90%) have less than 8 days of stock left as of the latest data — not a one-off, but a structural gap.
- 5 products are Critical: under 2 days of cover. These need action today, not at the next scheduled count.
- No dead stock was found , every product still sells , so the issue is under-stocking, not over-ordering the wrong things.
- Slowest-moving category: antibiotics (lowest stock turnover rate, 0.29) - cash is tied up here longer than in any other category.

Root Causes (Supported by Data)

What the numbers say is actually driving stockouts and shrinkage



Category loss concentration

Category	Loss (KES)
Fertilisers	2,997,200
Veterinary	310,460
Accessories	242,520
Seeds	163,780
All other (11 cats)	645,102

- Stock levels - not the day of week, month-end timing, or margin - are what predict a stockout. Low closing/opening stock and low days-of-cover are the clearest warning signs.
- Fertilisers alone cause ~69% of all shrinkage loss, driven by long 7-day supplier lead times, the longest of any category.
- Shrinkage is only weakly linked to margin (-0.24) — [uncertain] high-value items are not clearly "more stolen," so shrinkage looks more like handling/wastage than pure theft-of-value.

Possible Solutions

Three paths forward, from quick fixes to a full predictive system

1

Fix reorder points manually

Recalculate fixed reorder levels for the 18 low-cover products using the demand + lead-time formula already built. Fast, no new tooling.

2

Target fertiliser & agrovet supply chain

Renegotiate lead times or hold more safety stock for DAP/CAN Fertiliser and other agrovet lines - the single largest, fastest-payback fix.

3

Deploy the predictive risk model

Roll out the XGBoost model (99.9% ROC-AUC) to flag at-risk products days in advance and auto-suggest reorder quantities.

Trade-off: Option 1 is fastest but needs repeating manually; Option 2 delivers the biggest near-term KES saving; Option 3 scales best but takes longer to stand up and needs ongoing data feeds.

Recommended Solution & Expected Impact

Do all three, in sequence — starting with the biggest leak

Phase 1 (Now): Reorder the 5 Critical products

Product	Recommended qty	Est. cost (KES)
DAP Fertiliser 50kg	166	630,800
CAN Fertiliser 50kg	173	553,600
Drenching Gun Dewormer	102	48,960
Maize Seed 2kg	128	48,640
Newcastle Vaccine 100dos	105	26,250

Based on the reorder formula (7-day average demand x lead time + safety stock for a 95% service level). Full 20-product table is available on request.

Phase 2 & 3: Fix agroveter supply chain, then automate

- Renegotiate the 7-day fertiliser lead time or raise safety stock for agroveter lines.
- Pilot the risk model in agroveter locations first, then expand to all business types.

Model readiness

ROC-AUC (test set)	0.9993
Precision	0.98
Recall	0.99
F1 score	0.98

Expected impact

- Immediate: close the 5 Critical stock-outs before they cost sales.
- Near-term: reduce fertiliser/agroveter shrinkage, currently 69% of the KES 4.36M total loss.
- [uncertain] Ongoing: early-warning flags could materially cut the 2,918 stockout-risk events recorded, though the real-world reduction has not yet been measured in a live pilot.

Next Steps

A decision the stakeholder can act on this week

1

Approve

Sign off the Phase 1 reorder list (5 Critical products, ~KES 1.31M total) for immediate purchase.

2

Investigate

Review the 7-day fertiliser supplier lead time with agrovet suppliers to shorten the restock gap.

3

Pilot

Test the stockout-risk model on agrovet locations for one restock cycle before wider rollout.

4

Validate

[uncertain] Confirm findings against live sales data — this analysis used an engineered dataset built to reflect realistic conditions, not verified historical records.